Chapter 3
Memory Map

3.1 Introduction

This chip contains various memories and memory-mapped peripherals that are located in one 32-bit contiguous memory space. This chapter describes the memory and peripheral locations within that memory space.

Details about the memory map appear in the spreadsheet that is attached to this document: S32K1xx_memory_map.xlsx. To access this spreadsheet, view the document's list of attachments.

3.2 SRAM memory map

3.2.1 S32K14x: SRAM memory map

The on-chip RAM is split in two regions: SRAM_L and SRAM_U. The RAM is implemented such that the SRAM_L and SRAM_U ranges form a contiguous block in the memory map. See S32K1xx_memory_map.xlsx attached to this document for details.

Accesses to the SRAM_L and SRAM_U memory ranges outside the amount of RAM on the chip causes the bus cycle to be terminated with an error followed by the appropriate response in the requesting bus master.

3.2.2 S32K11x: SRAM memory map

In S32K11x on-chip RAM can be used in following applications:

- Safety critical applications: SRAM_U can be used, which starts from 2000_0000
- Non-safety critical applications: SRAM_U along with 1 KB MTB can be used

NOTE

CM0+ architecture implements single RAM controller, hence SRAM_U and MTB are referred as single contiguous memory regions. MTB in S32K11x is present at the same location as SRAM_L in S32K14x.

See S32K1xx_memory_map.xlsx attached to this document for details. Accesses to the memory ranges outside the amount of RAM on the chip causes the bus cycle to be terminated with an error followed by the appropriate response in the requesting bus master.

3.3 Flash memory map

The various flash memories and the flash memory registers are located at different base addresses as shown in the following figure. The base address for each is specified in the S32K1xx_memory_map.xlsx file attached to this document.

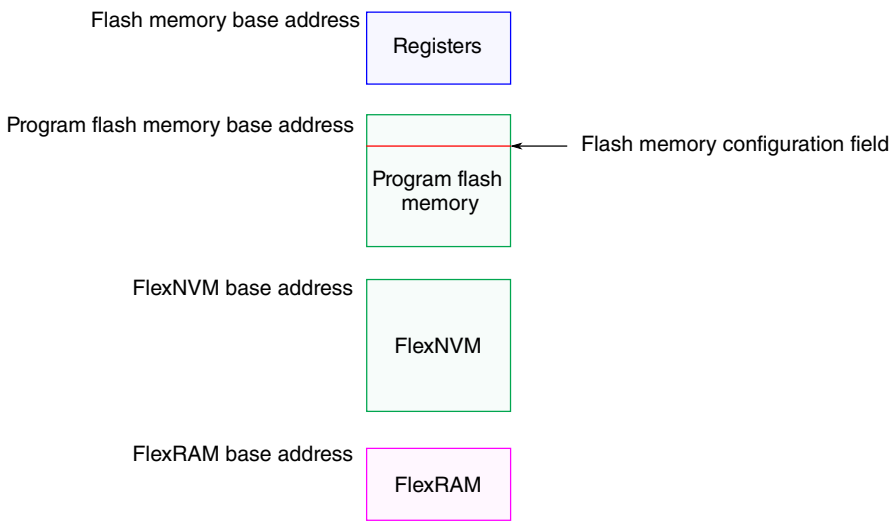


Figure 3-1. Flash memory map

3.4 Peripheral bridge (AIPS-Lite) memory map

The peripheral memory map is accessible via a crossbar slave port.

There are three regions associated with peripheral space, as shown in the following table.

Table 3-1. Regions associated with peripheral space

Address space	Region description
0x4000_0000–0x4001_FFFF	A 128 KB region, partitioned as 32 spaces, each 4 KB in size and reserved for on-platform peripheral devices. AIPS-Lite generates unique module enables for all 32 spaces.
0x4002_0000–0x4007_FFFF	A 384 KB region, partitioned as 96 spaces, each 4 KB in size and reserved for off-platform modules. AIPS-Lite generates unique module enables for all 96 spaces.
0x400F_F000	A 4 KB region for accessing the GPIO module. This block is connected to the AMBA bus via the port splitter and provides direct master access without incurring wait states associated with accesses via the AIPS-Lite modules. The GPIO is implemented only in the upper space of this region (4 KB beginning at 0x400F_F000).

Modules that are disabled via their clock gate control bits in the PCC/SIM registers disable the associated AIPS-Lite slots. Access to any address within an unimplemented or disabled peripheral bridge slot results in a transfer error termination.

NOTE

While trying to access memory map region of unavailable feature (See SIM_SDID[FEATURES]) with corresponding module clock enabled through PCC CGC bit, there will not be any transfer error termination.

3.4.1 Read-after-write sequence and required serialization of memory operations

In some situations, a write to a peripheral must be completed fully before a subsequent action can occur. Examples of such situations include:

- Exiting an interrupt service routine (ISR)
- Changing a mode
- Configuring a function

In these situations, the application software must perform a read-after-write sequence to guarantee the required serialization of the memory operations, as shown in the following table.

Table 3-2. Read-after-write sequence to guarantee required serialization of memory operations

Step	Action
1	Write the peripheral register.
2	Read the written peripheral register to verify the write.
3	Continue with subsequent operations.

NOTE

In S32K14x series of devices, one factor contributing to these situations is processor write buffering. The processor architecture has a programmable configuration field to disable write buffering: ACTLR[DISDEFWBUF]. (For details see Arm® Cortex® M4 Processor Technical Reference Manual, Revision r0p1, at http://arm.com). However, disabling buffered writes is likely to degrade system performance much more than simply performing the required memory serialization for the situations that truly require it.

3.5 Private Peripheral Bus (PPB) memory map

The PPB is part of the defined Arm bus architecture and provides access to select processor-local modules. These resources are only accessible from the core; other system masters do not have access to them.

Table 3-3. PPB memory map for CM4

System 32-bit address range	Resource
0xE000_0000–0xE000_0FFF	Instrumentation Trace Macrocell (ITM)
0xE000_1000–0xE000_1FFF	Data Watchpoint and Trace (DWT)
0xE000_2000–0xE000_2FFF	Flash Patch and Breakpoint (FPB)
0xE000_3000–0xE000_DFFF	Reserved
0xE000_E000–0xE000_EFFF	System Control Space (SCS) (for NVIC and FPU
0xE000_F000–0xE003_FFFF	Reserved
0xE004_0000–0xE004_0FFF	Trace Port Interface Unit (TPIU)
0xE004_1000–0xE004_1FFF	Reserved
0xE004_2000–0xE004_2FFF	Reserved
0xE004_3000–0xE004_3FFF	Reserved
0xE004_4000–0xE007_FFFF	Reserved
0xE008_0000–0xE008_0FFF	Miscellaneous Control Module (MCM)
0xE008_1000–0xE008_1FFF	Reserved
0xE008_2000–0xE008_2FFF	Cache Controller (LMEM)
0xE008_3000–0xE00F_EFFF	Reserved
0xE00F_F000–0xE00F_FFFF	Arm Core ROM Table[1] - allows auto-detection of debug components

1. The Arm Core ROM table is optionally required by Arm CoreSight debug infrastructure to discover the components on the chip. This ROM table has no any relationship with the MCU Boot ROM.

Table 3-4. PPB memory map for CM0+

System 32-bit address range	Resource
0xE000_0000–0xE000_0FFF	Reserved [1]
0xE000_1000–0xE000_1FFF	Data Watchpoint and Trace (DWT)
0xE000_2000–0xE000_2FFF	Arm Core ROM Table
0xE000_3000–0xE000_DFFF	Reserved[1]
0xE000_E000–0xE000_EFFF	System Control Space (SCS) (for NVIC)
0xE000_F000–0xE003_FFFF	Reserved[1]
0xE004_0000–0xE004_0FFF	Reserved[1]
0xE004_1000–0xE004_1FFF	Reserved[1]
0xE004_2000–0xE004_2FFF	Reserved[1]
0xE004_3000–0xE004_3FFF	Reserved[1]
0xE004_4000–0xE007_FFFF	Reserved[1]
0xE008_0000–0xE008_0FFF	Reserved[1]
0xE008_1000–0xE008_1FFF	Reserved[1]
0xE008_2000–0xE008_2FFF	Reserved[1]
0xE008_3000–0xE00F_EFFF	Reserved[1]
0xE00F_F000–0xE00F_FFFF	Reserved[1]
0xE010_0000-0xEFFF_FFFF	Reserved[1]

1. Reserved area return transfer error. Reserved is Unknown/Should be 0. Software must not rely on reading as all 0s, and it must treat the value as if it is unknown.

3.6 Aliased bit-band regions for CM4 core

The SRAM_U, AIPS-Lite, and general purpose input/output (GPIO) module resources reside in the Cortex-M4F processor bit-band regions.

The processor also includes two 32 MB aliased bit-band regions associated with the two 1 MB bit-band spaces. Each 32-bit location in the 32 MB space maps to an individual bit in the bit-band region. A 32-bit write in the alias region has the same effect as a read-modify-write operation on the targeted bit in the bit-band region.

Bit 0 of the value written to the alias region determines what value is written to the target bit:

- Writing a value with bit 0 set writes a 1 to the target bit.
- Writing a value with bit 0 clear writes a 0 to the target bit.

A 32-bit read in the alias region returns either:

- a value of 0x0000_0000 to indicate the target bit is clear
- a value of 0x0000_0001 to indicate the target bit is set

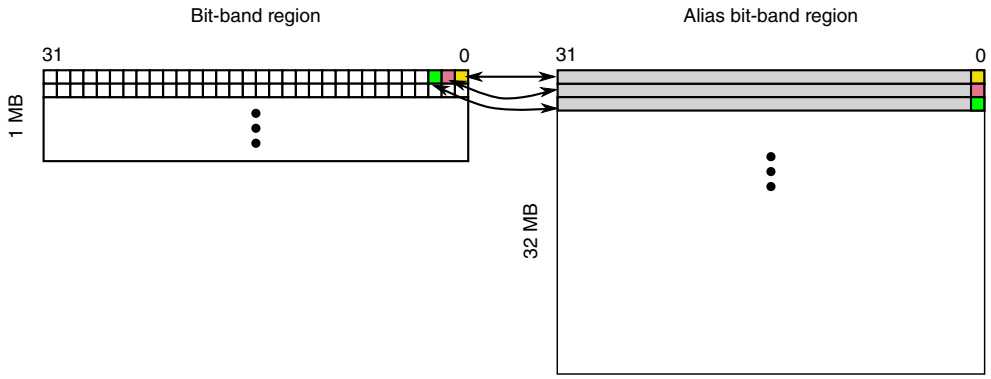


Figure 3-2. Alias bit-band mapping

NOTE

Each bit in a bit-band region has an equivalent bit that can be manipulated through bit 0 in a corresponding long word in the alias bit-band region.

NOTE

Do not use bit banding for w1c status bits.

CAUTION

The S32K product series and the software drivers support bit-banding, but Arm no longer promotes its usage. Therefore, we recommend that bit-banding should not be used.